

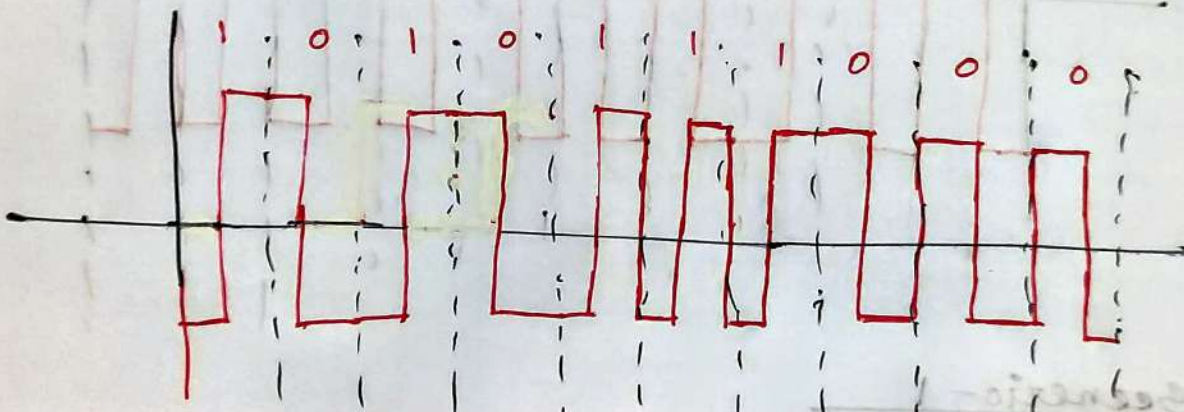
Q₁
Signal A → 1010111000

Signal B → 1110111011

AND 1010111000

No consecutive zero problem : use MLT-3 / Manchester / Differential Manchester

using Manchester



Q₂

3 bit → 4 bit

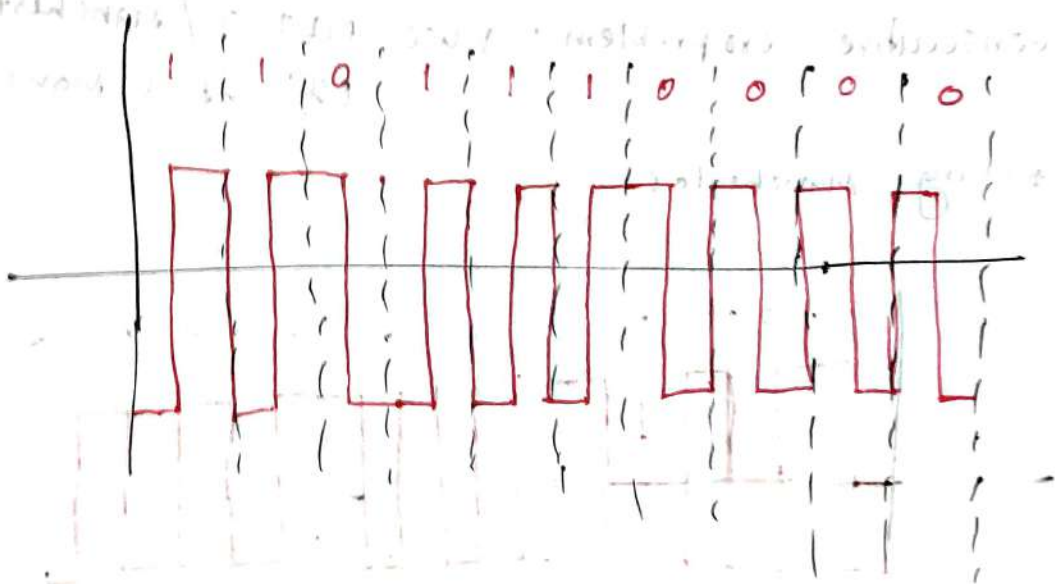
↓
 $2^3 = 8$ combination

3 bit	4 bit
111	0010
110	0011
101	0100
100	0101
011	0110
010	1001
001	1010
000	1011

Q3

① 1101110000

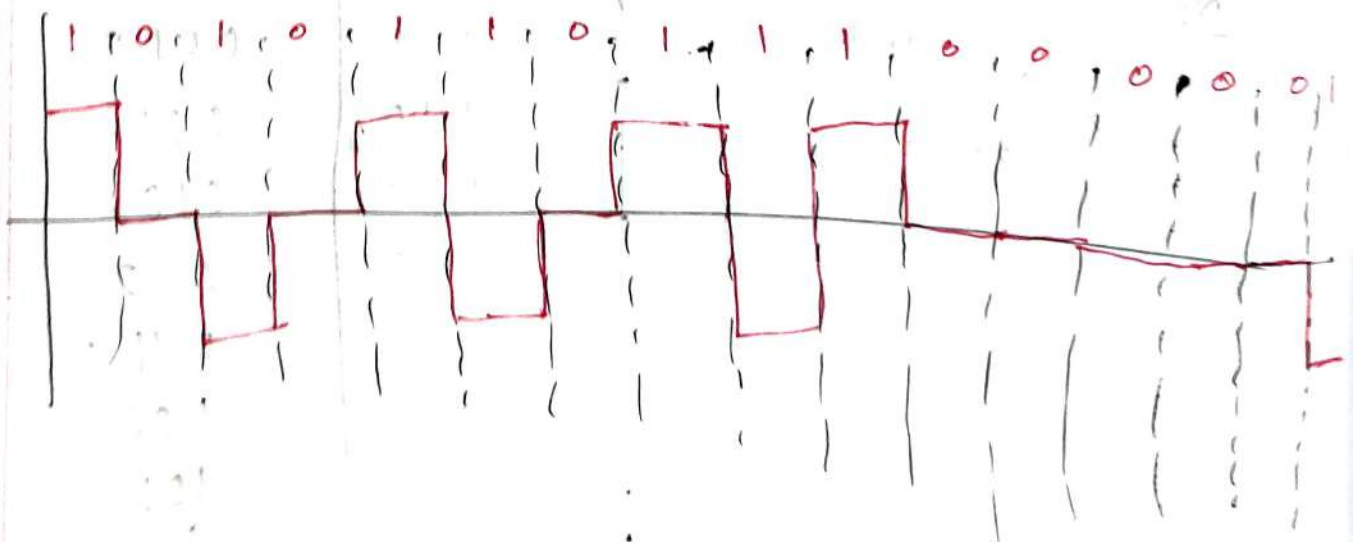
②



Q4

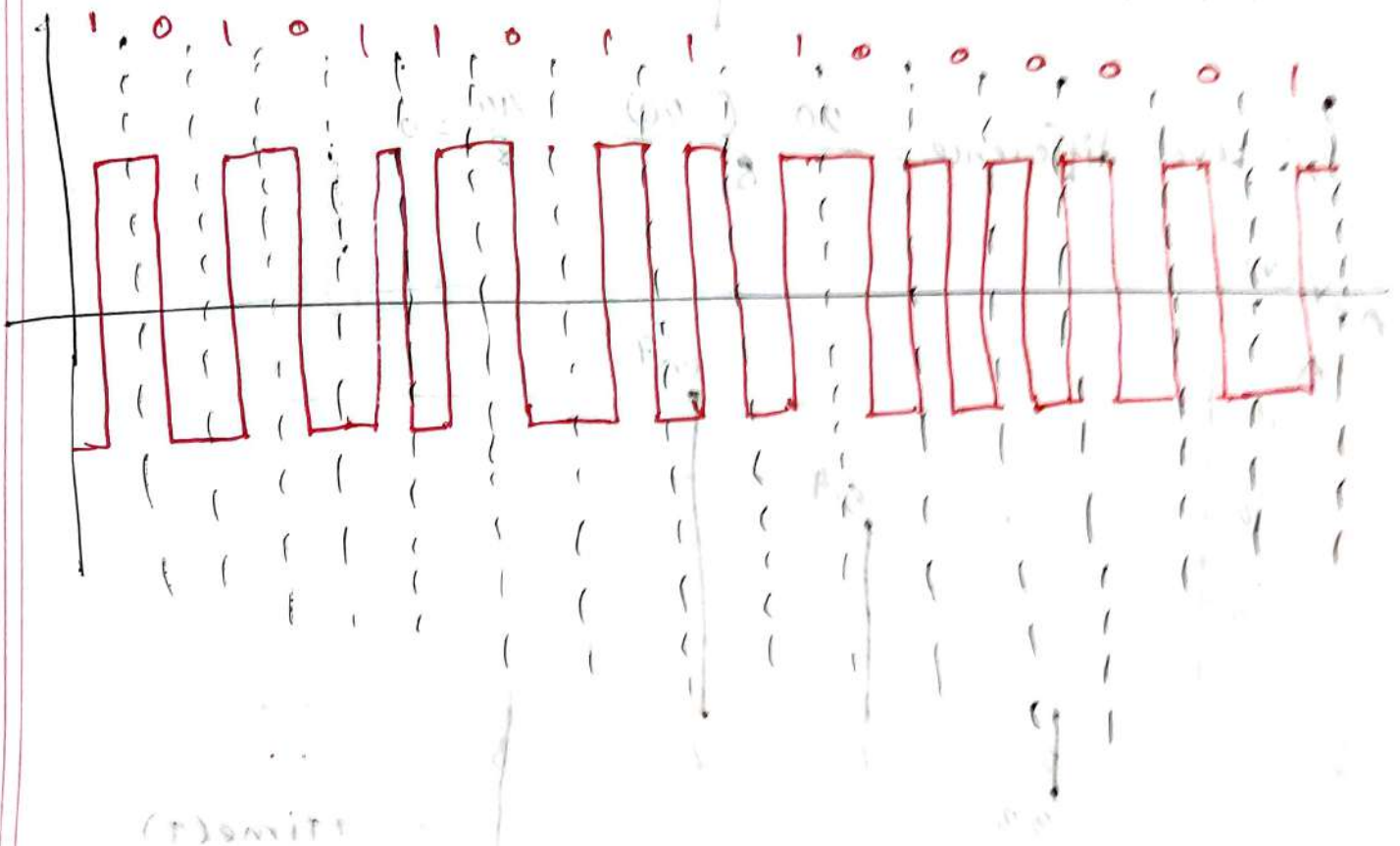
Scenario-1

AMI $\rightarrow$ no Base line Wandering, low broadband



Scenario-2

Use Manchester / Differential Manchester



(1) transmit

0.07

10.2	11.2	12.2	13.2
2.2	2.2	2.2	2.2
11.2	12.2	13.2	14.2
15.2	16.2	17.2	18.2
19.2	20.2	21.2	22.2
23.2	24.2	25.2	26.2
27.2	28.2	29.2	30.2
31.2	32.2	33.2	34.2
35.2	36.2	37.2	38.2
39.2	40.2	41.2	42.2
43.2	44.2	45.2	46.2
47.2	48.2	49.2	50.2
51.2	52.2	53.2	54.2
55.2	56.2	57.2	58.2
59.2	60.2	61.2	62.2
63.2	64.2	65.2	66.2
67.2	68.2	69.2	70.2
71.2	72.2	73.2	74.2
75.2	76.2	77.2	78.2
79.2	80.2	81.2	82.2
83.2	84.2	85.2	86.2
87.2	88.2	89.2	90.2
91.2	92.2	93.2	94.2
95.2	96.2	97.2	98.2
99.2	100.2	101.2	102.2
103.2	104.2	105.2	106.2
107.2	108.2	109.2	110.2
111.2	112.2	113.2	114.2
115.2	116.2	117.2	118.2
119.2	120.2	121.2	122.2
123.2	124.2	125.2	126.2
127.2	128.2	129.2	130.2
131.2	132.2	133.2	134.2
135.2	136.2	137.2	138.2
139.2	140.2	141.2	142.2
143.2	144.2	145.2	146.2
147.2	148.2	149.2	150.2
151.2	152.2	153.2	154.2
155.2	156.2	157.2	158.2
159.2	160.2	161.2	162.2
163.2	164.2	165.2	166.2
167.2	168.2	169.2	170.2
171.2	172.2	173.2	174.2
175.2	176.2	177.2	178.2
179.2	180.2	181.2	182.2
183.2	184.2	185.2	186.2
187.2	188.2	189.2	190.2
191.2	192.2	193.2	194.2
195.2	196.2	197.2	198.2
199.2	200.2	201.2	202.2
203.2	204.2	205.2	206.2
207.2	208.2	209.2	210.2
211.2	212.2	213.2	214.2
215.2	216.2	217.2	218.2
219.2	220.2	221.2	222.2
223.2	224.2	225.2	226.2
227.2	228.2	229.2	230.2
231.2	232.2	233.2	234.2
235.2	236.2	237.2	238.2
239.2	240.2	241.2	242.2
243.2	244.2	245.2	246.2
247.2	248.2	249.2	250.2
251.2	252.2	253.2	254.2
255.2	256.2	257.2	258.2
259.2	260.2	261.2	262.2
263.2	264.2	265.2	266.2
267.2	268.2	269.2	270.2
271.2	272.2	273.2	274.2
275.2	276.2	277.2	278.2
279.2	280.2	281.2	282.2
283.2	284.2	285.2	286.2
287.2	288.2	289.2	290.2
291.2	292.2	293.2	294.2
295.2	296.2	297.2	298.2
299.2	300.2	301.2	302.2
303.2	304.2	305.2	306.2
307.2	308.2	309.2	310.2
311.2	312.2	313.2	314.2
315.2	316.2	317.2	318.2
319.2	320.2	321.2	322.2
323.2	324.2	325.2	326.2
327.2	328.2	329.2	330.2
331.2	332.2	333.2	334.2
335.2	336.2	337.2	338.2
339.2	340.2	341.2	342.2
343.2	344.2	345.2	346.2
347.2	348.2	349.2	350.2
351.2	352.2	353.2	354.2
355.2	356.2	357.2	358.2
359.2	360.2	361.2	362.2
363.2	364.2	365.2	366.2
367.2	368.2	369.2	370.2
371.2	372.2	373.2	374.2
375.2	376.2	377.2	378.2
379.2	380.2	381.2	382.2
383.2	384.2	385.2	386.2
387.2	388.2	389.2	390.2
391.2	392.2	393.2	394.2
395.2	396.2	397.2	398.2
399.2	400.2	401.2	402.2
403.2	404.2	405.2	406.2
407.2	408.2	409.2	410.2
411.2	412.2	413.2	414.2
415.2	416.2	417.2	418.2
419.2	420.2	421.2	422.2
423.2	424.2	425.2	426.2
427.2	428.2	429.2	430.2
431.2	432.2	433.2	434.2
435.2	436.2	437.2	438.2
439.2	440.2	441.2	442.2
443.2	444.2	445.2	446.2
447.2	448.2	449.2	450.2
451.2	452.2	453.2	454.2
455.2	456.2	457.2	458.2
459.2	460.2	461.2	462.2
463.2	464.2	465.2	466.2
467.2	468.2	469.2	470.2
471.2	472.2	473.2	474.2
475.2	476.2	477.2	478.2
479.2	480.2	481.2	482.2
483.2	484.2	485.2	486.2
487.2	488.2	489.2	490.2
491.2	492.2	493.2	494.2
495.2	496.2	497.2	498.2
499.2	500.2	501.2	502.2
503.2	504.2	505.2	506.2
507.2	508.2	509.2	510.2
511.2	512.2	513.2	514.2
515.2	516.2	517.2	518.2
519.2	520.2	521.2	522.2
523.2	524.2	525.2	526.2
527.2	528.2	529.2	530.2
531.2	532.2	533.2	534.2
535.2	536.2	537.2	538.2
539.2	540.2	541.2	542.2
543.2	544.2	545.2	546.2
547.2	548.2	549.2	550.2
551.2	552.2	553.2	554.2
555.2	556.2	557.2	558.2
559.2	560.2	561.2	562.2
563.2	564.2	565.2	566.2
567.2	568.2	569.2	570.2
571.2	572.2	573.2	574.2
575.2	576.2	577.2	578.2
579.2	580.2	581.2	582.2
583.2	584.2	585.2	586.2
587.2	588.2	589.2	590.2
591.2	592.2	593.2	594.2
595.2	596.2	597.2	598.2
599.2	600.2	601.2	602.2
603.2	604.2	605.2	606.2
607.2	608.2	609.2	610.2
611.2	612.2	613.2	614.2
615.2	616.2	617.2	618.2
619.2	620.2	621.2	622.2
623.2	624.2	625.2	626.2
627.2	628.2	629.2	630.2
631.2	632.2	633.2	634.2
635.2	636.2	637.2	638.2
639.2	640.2	641.2	642.2
643.2	644.2	645.2	646.2
647.2	648.2	649.2	650.2
651.2	652.2	653.2	654.2
655.2	656.2	657.2	658.2
659.2	660.2	661.2	662.2
663.2	664.2	665.2	666.2
667.2	668.2	669.2	670.2
671.2	672.2	673.2	674.2
675.2	676.2	677.2	678.2
679.2	680.2	681.2	682.2
683.2	684.2	685.2	686.2
687.2	688.2	689.2	690.2
691.2	692.2	693.2	694.2
695.2	696.2	697.2	698.2
699.2	700.2	701.2	702.2
703.2	704.2	705.2	706.2
707.2	708.2	709.2	710.2
711.2	712.2	713.2	714.2
715.2	716.2	717.2	718.2
719.2	720.2	721.2	722.2
723.2	724.2	725.2	726.2
727.2	728.2	729.2	730.2
731.2	732.2	733.2	734.2
735.2	736.2	737.2	738.2
739.2	740.2	741.2	742.2
743.2	744.2	745.2	746.2
747.2	748.2	749.2	750.2
751.2	752.2	753.2	754.2
755.2	756.2	757.2	758.2
759.2	760.2	761.2	762.2
763.2	764.2	765.2	766.2
767.2	768.2	769.2	770.2
771.2	772.2	773.2	774.2
775.2	776.2	777.2	778.2
779.2	780.2	781.2	782.2
783.2	784.2	785.2	786.2
787.2	788.2	789.2	790.2
791.2	792.2	793.2	794.2
795.2	796.2	797.2	798.2
799.2	800.2	801.2	802.2
803.2	804.2	805.2	806.2
807.2	808.2	809.2	810.2
811.2	812.2	813.2	814.2
815.2	816.2	817.2	818.2
819.2	820.2	821.2	822.2
823.2	824.2	825.2	826.2
827.2	828.2	829.2	830.2
831.2	832.2	833.2	834.2
835.2	836.2	837.2	838.2
839.2	840.2	841.2	842.2
843.2	844.2	845.2	846.2
847.2	848.2	849.2	850.2
851.2	852.2	853.2	854.2
855.2	856.2	857.2	858.2
859.2	860.2	861.2	862.2
863.2	864.2	865.2	866.2
867.2	868.2	869.2	870.2
871.2	872.2	873.2	874.2
875.2	876.2	877.2	878.2
879.2	880.2	881.2	882.2
883.2	884.2	885.2	886.2
887.2	888.2	889.2	890.2
891.2	892.2	893.2	894.2
895.2	896.2	897.2	898.2
899.2	900.2	901.2	902.2
903.2	904.2	905.2	906.2
907.2	908.2	909.2	910.2
911.2	912.2	913.2	914.2
915.2	916.2	917.2	918.2
919.2	920.2	921.2	922.2
923.2	924.2	925.2	926.2
927.2	928.2	929.2	930.2
931.2	932.2	933.2	934.2
935.2	936.2	937.2	938.2
939.2	940.2	941.2	942.2
943.2	944.2	945.2	946.2
947.2	948.2	949.2	950.2
951.2	952.2	953.2	954.2
955.2	956.2	957.2	958.2
959.2	960.2	961.2	962.2
963.2	964.2	965.2	966.2
967.2	968.2	969.2	970.2
971.2	972.2	973.2	974.2
975.2	976.2	977.2	978.2
979.2	980.2	981.2	982.2
983.2	984.2	985.2	986.2
987.2	988.2	989.2	990.2
991.2	992.2	993.2	994.2
995.2	996.2	997.2	998.2
999.2	1000.2	1001.2	1002.2
1003.2	1004.2	1005.2	1006.2
1007.2	1008.2	1009.2	1010.2
1011.2	1012.2	1013.2	1014.2
1015.2	1016.2	1017.2	1018.2
1019.2	1020.2	1021.2	1022.2
1023.2	1024.2	1025.2	1026.2
1027.2	1028.2	1029.2	1030.2
1031.2	1032.2	1033.2	1034.2
1035.2	1036.2	1037.2	1038.2
1039.2	1040.2	1041.2	1042.2
1043.2	1044.2	1045.2	1046.2
1047.2	1048.2	1049.2	1050.2
1051.2	1052.2	1053.2	1054.2
1055.2	1056.2	1057.2	1058.2
1059.2	1060.2	1061.2	1062.2
1063.2	1064.2	1065.2	1066.2

Q5

2-0189N330

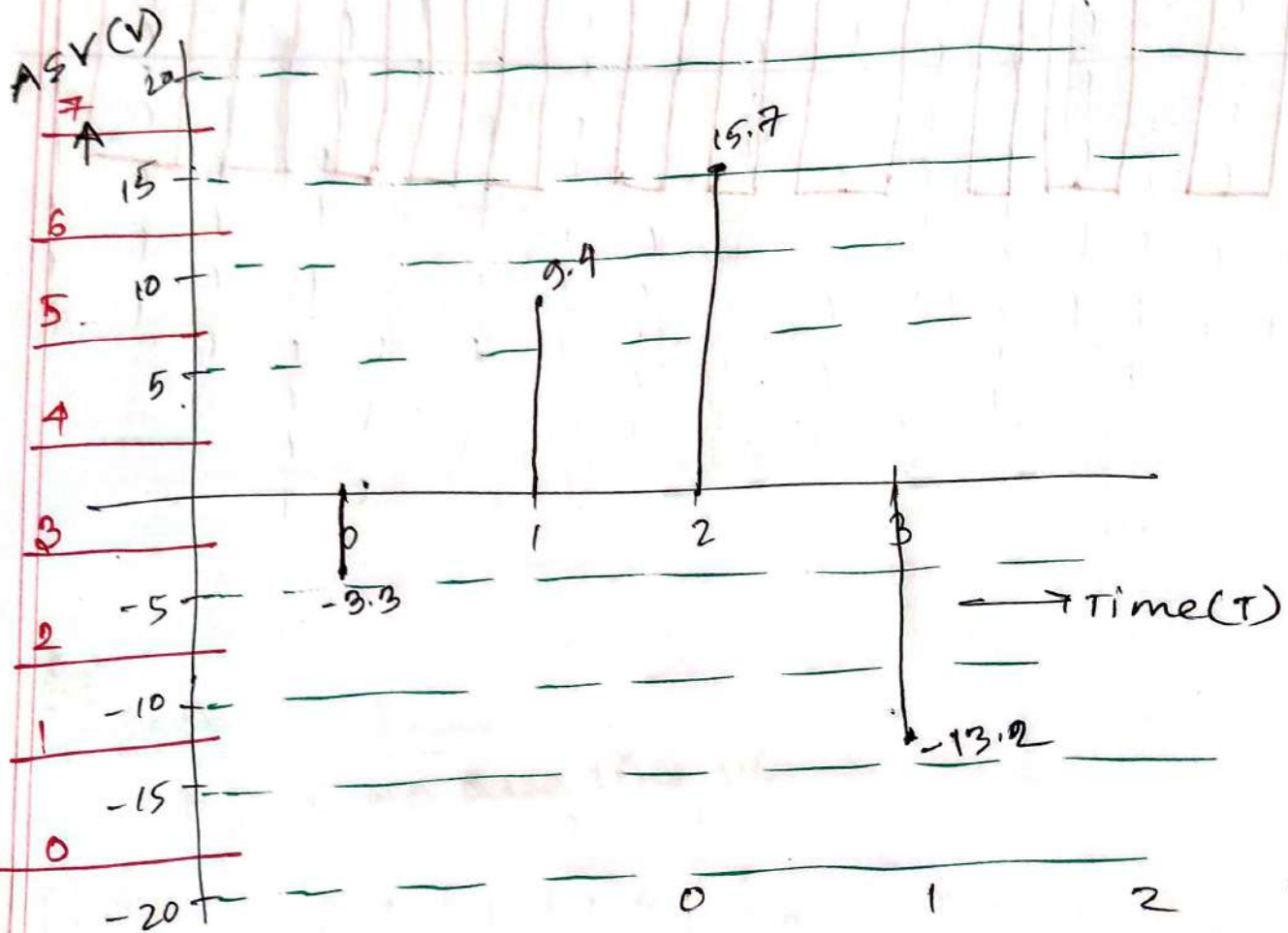
Max Amplitude = 20 V

Min Amplitude = -20 V

of bits = 3

$\therefore \text{level} = 2^3 = 8$

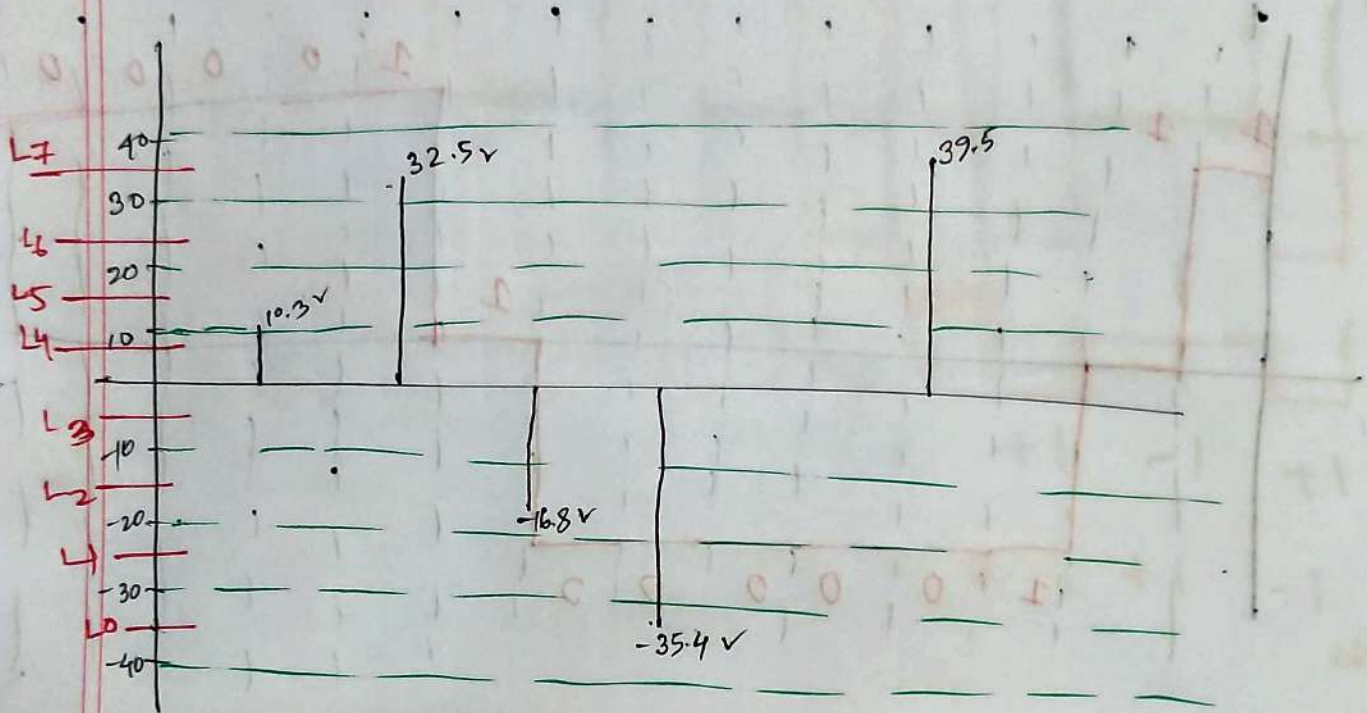
$$\therefore \text{Level Difference} = \frac{20 - (-20)}{8} = \frac{40}{8} = 5$$



Normalized PMA value	-0.66	1.88	3.14	-2.64
Normalized quantized value	-0.5	1.5	3.5	-2.5
Error	0.16	-0.38	0.36	0.14
Quantized code	1	2	4	3
Encoded value	001	010	100	011

Q6

Level = $2^3 = 8$ | Level Difference = $\frac{80}{8} = 10$



	1	2	3	4	5
Normalized PAM value	1.2875	4.0625	-2.1	-4.425	4.9375
Normalized quantized value	1.5	3.5	-1.5	-3.5	3.5
Quantization code	2	4	2	4	4
Encoded value	010	100	010	100	100

0 1 0 0 0 1 0 0 0 0 0 0 0 1 1 1 1

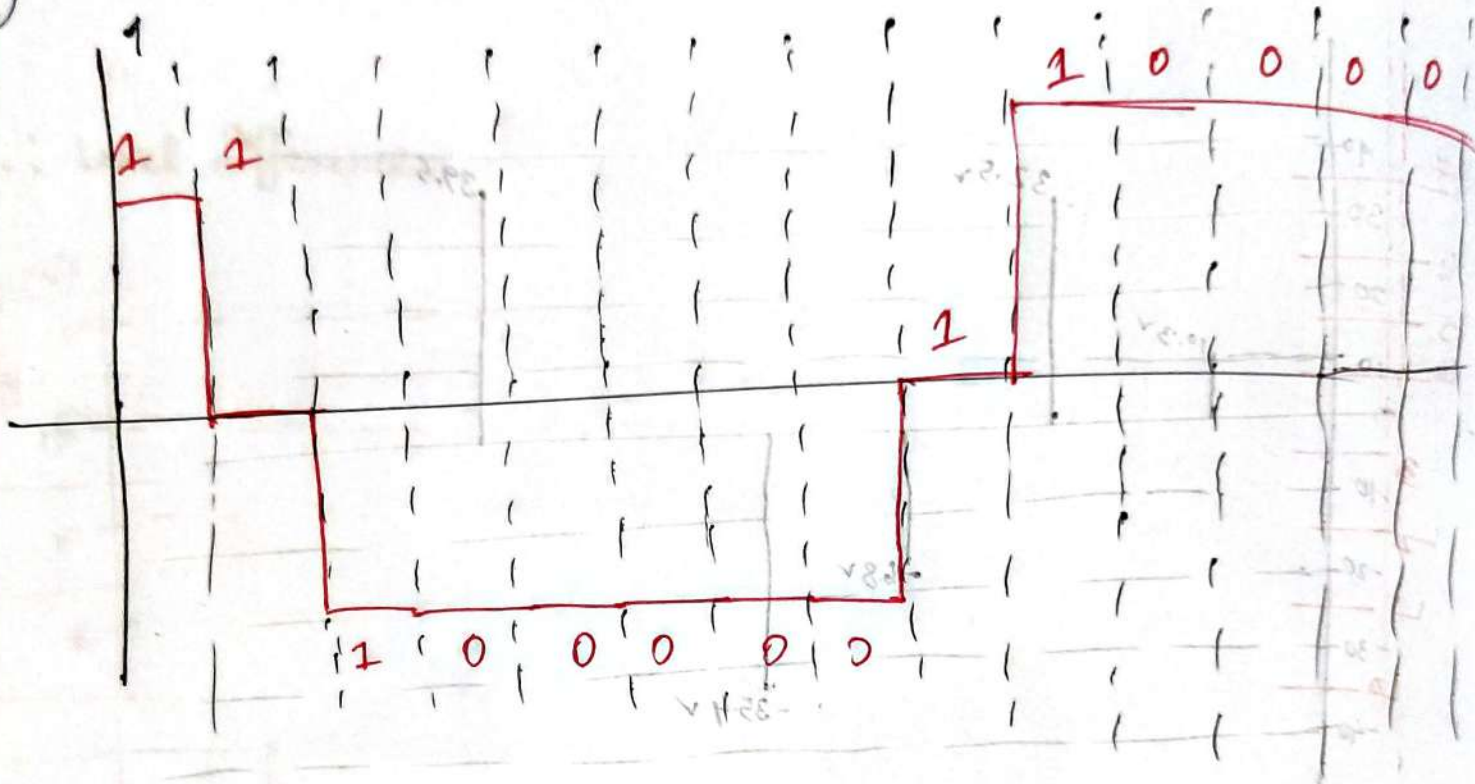


Q-8

MLT-3

Level = 8, Level Difference = 8

(i)



(ii)

Baseline wandering → consecutive 0.

DC component → No

(iii)

HDB3

EVEN

8 0 0 0 V

